

Fluid Flow in Pipes

Pre-Lab Plan
Unit Operations Lab 4
10/15/2019

Group 1, Tuesday, AM section

Team member Names:

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1. The Experimental Objective:

The purpose of this experiment is to obtain the following energy loss coefficients which are friction factor (f), loss coefficient (K) and the flow coefficient (C_f) on different sections of fluid flow system. Furthermore, comparing the obtained data with the data from theories is another purpose of this experiment in order to establish the correlation between fluid flow in pipes and the fitting.

2. Experimental:

➤ 2.1 Apparatus

Figure 1: The most important components used in this experiment.

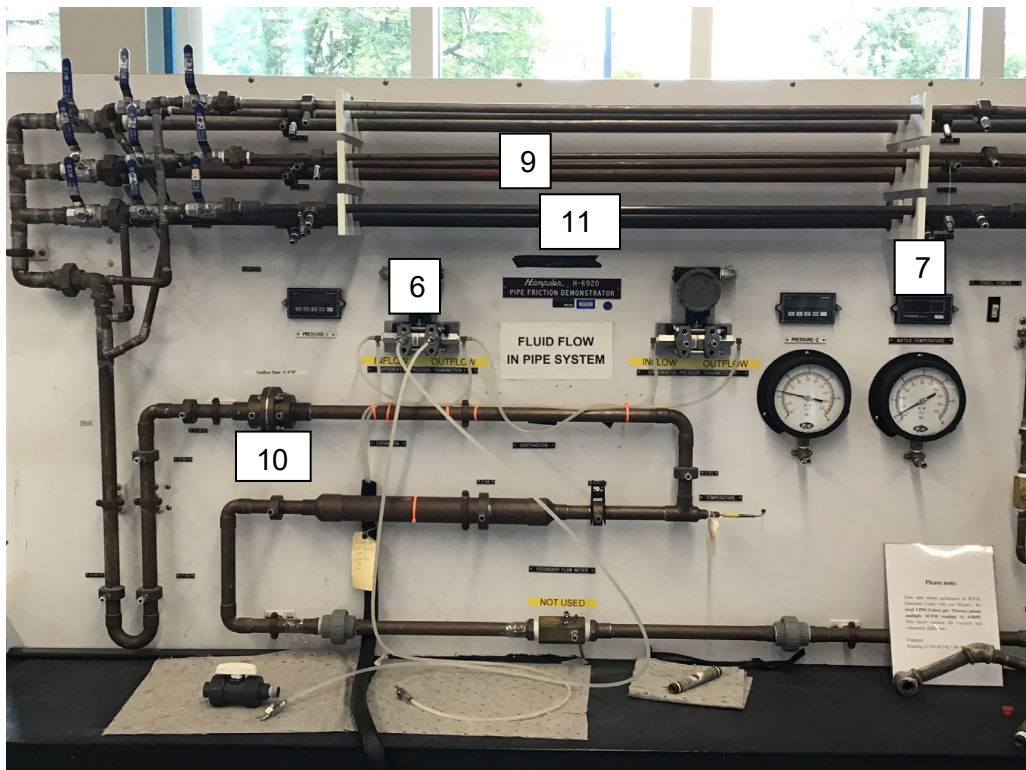


Figure 2: The most important components used in this experiment.

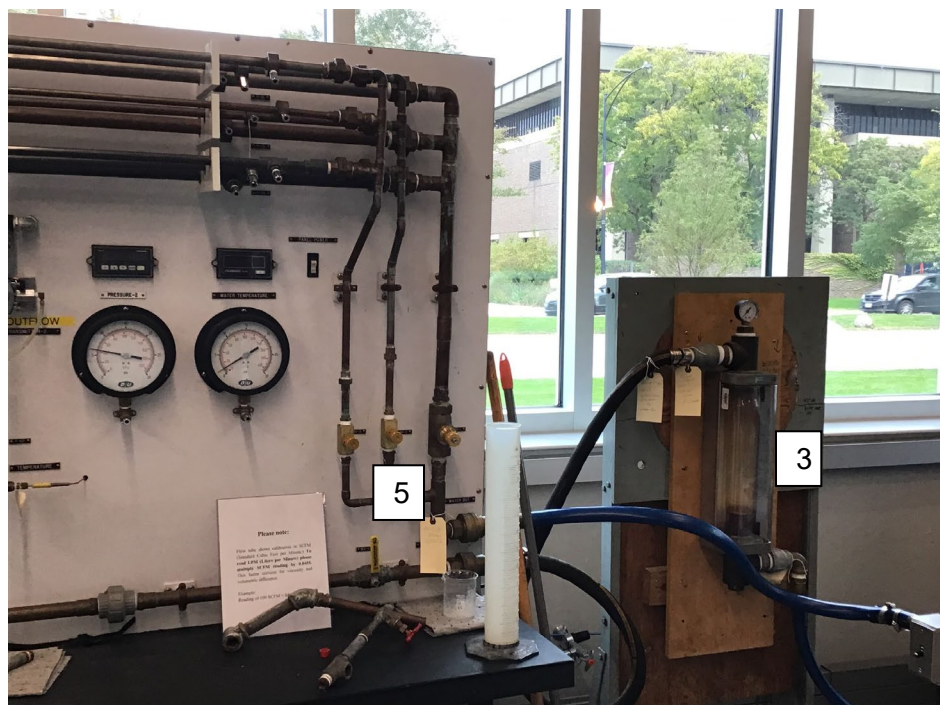


Figure 3: The most important components used in this experiment.

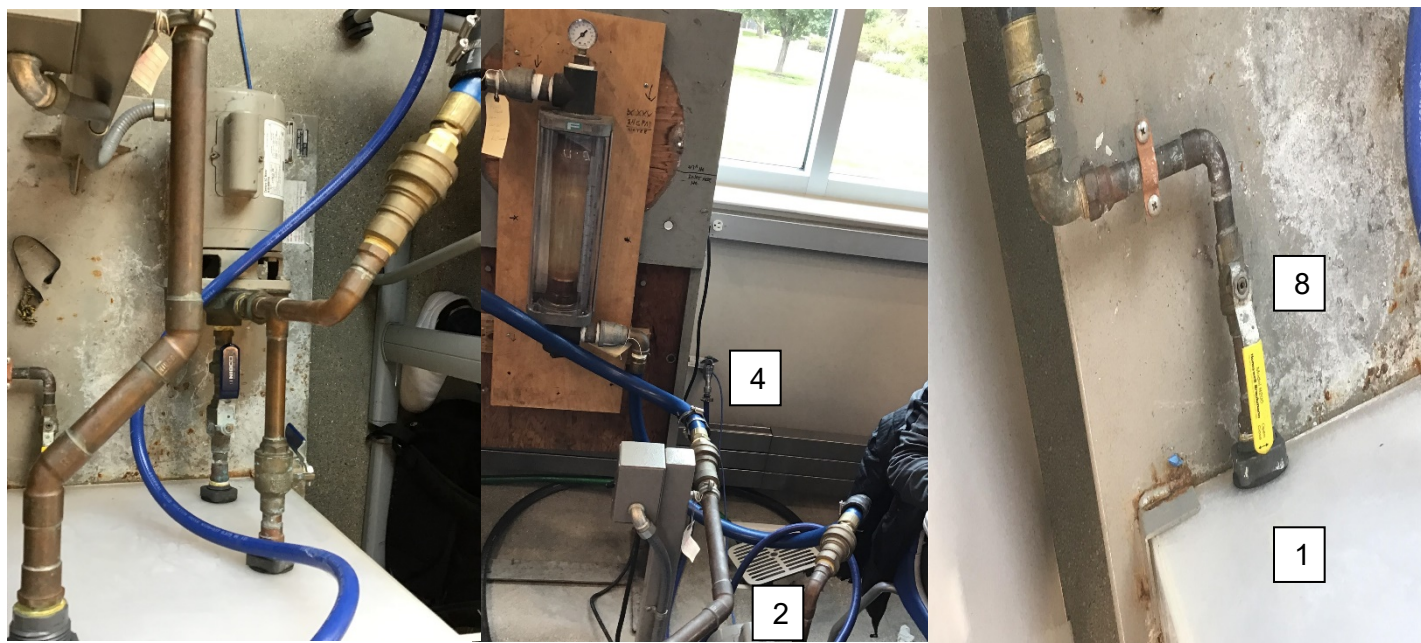


Table 1: The most important components used in this experiment

#	The important components	Description
1	reservoir tank	Used to hold water that will be used and pushed into the system.
2	pump	Used to push the water into the system.
3	rotameter	Used to control the flow of water into and out of the system
4	The shut-off valves	Used in order to create different setup of the piping system by manually opening / closing these valves.
5	The flow control valves	Used to control the flow of water into the system manually.
6	Differential Pressure transmitter	Digital device used to measure the pressure drop in the system by measuring the pressure in and pressure out.
7	Digital thermostat	Digital device used to measure the temperature in the system
8	Water release valve	Valve that releases the water to the drain
9	Flow Circuit	Series of pipes that are made of different pipe materials which are: Copper tubing, Plastic piping and Stainless Steel piping
10	Orifice flowmeter	Used to measure the differential pressure across the orifice plate
11	Thin Plastic tubes with	Used to connect the pressure gauges with the certain location of the

	quick-disconnect fittings at each end	piping/fitting in order to measure the pressure drop
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➤ 2.2 Materials and Supplies

Table 2: The material and supplies used in this experiment

#	Material and Supplies	Description
1	Water	Tap water used in pipes
2	Bucket	Used to calibrate the rotameter
3	Rag	Used to clean up spills
4	Timer	Used for the flow rate calibration
5	Ruler	Used to measure pipe lengths
6	Graduated cylinder	Used to measure the volume of the fluid (Water)

➤ 2.3 Experimental Plan

1. Rotameter Calibration:

- Fill the tank halfway with water by opening the valve and switching on the pump
- Find the correct conversion rate from SCFM to LPM by collecting flow in a large bucket for a set time. Then pour the water into a graduated cylinder to find the flow rate in LPM. Compare this to the reading from the flow meter.
- Repeat for 3 different flow rates (5,10,15 LPM) to see deviations.

2. Orifice Meter:

- Check if DPT is working correctly. If while unconnected it is not reading close to 0 (0.00, 0.002), call the instructor to possibly re-zero. Avoid using manual Dp readers.
- Connect DPT tubes across the orifice.
- Determine C_f by allowing flow through main pipe.
- Record water temperature and pressure across orifice for each flow rate.
- Repeat process for a few flow rates. Calculate C_f .

3. Pipe Friction Head Loss:

- A. Determine friction factor as a function of Reynolds number by measuring the pressure drop over the pipe across different flow rates (5,10,15 LPM) and different pipe types (PVC, copper, and stainless steel).
- B. Compare friction factor to empirically and published data.
- C. Compare fanning friction factor and darcy friction factor
- D. Repeat across the 3 pipe sizes.

4. Pipe Fitting Head Loss:

- A. Determine the loss coefficient for the different fittings (bends, elbows, expansion, and contraction) as a function of Reynolds number.
- B. Measure pressure drop, flow rate, and water temperature across the different fittings.
- C. Find the fanning friction factor and the length of each fitting.

5. Pressure Drop Measurements:

- A. Do this every time a new orifice, pipe, or fitting needs to be measured.
- B. Close valve to stop water flow
- C. Pull out the quick disconnect fittings from the sensing ports.
- D. Connect into sensing ports into the next section to be measured, and open the valve to start flow.
- E. Flush out DPT by opening the two back valves.
- F. After flow has become steady, read the differential pressure on the digital interface.

Error can be reduced by doing every reading twice.

The independent variables in this experiment are the types of fittings, types/size of pipe, and flow rate.

The dependent variable is the differential pressure and the water temperature.

Standard deviation will be taken across these measurements. The confidence interval will be 95%



Project Safety Evaluation (10 pts)

Project Title: Fluid Flow in Pipes

Date: 10/15/19

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: There are electrical thermometers & outlets that need to stay dry
Potential mechanical hazards?	Yes	If yes, identify conditions: If there is no water in the pump, the pump can be damaged
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	Yes	If yes, specify type: latex lab gloves
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: water (7732-18-5)
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	25°C	
Pressure (psia)	25	Should be throughout the system
Safety Controls	yes/no	If yes, Provide description
	Yes	water shut off valve & water release valve used if water is accumulation occurs

I have read relevant background material for the Unit Operations Laboratory entitled: "Fluid Flow in Pipe" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

signature of team member [Signature] date 10/15/19
signature of team member [Signature] date 10/15/19
signature of team member [Signature] date 10/15/19